

Andrei-Cătălin Popa

EMJM STUDENT IN ASTROPHYSICS AND SPACE SCIENCE

🏠 acpopa.github.io

Education

Université Côte d'Azur

Nice, France

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Mar. 2025 – Sept. 2026

Thesis: *Characterising M-dwarf Stars as Exoplanet Hosts in Preparation for the PLATO Mission.*

Advisors: Dr. Orlagh Creevey (OCA), Dr. Valentina D'Orazi (Tor Vergata), Dr. Ulrike Heiter (Uppsala).

Research Internship: *Fundamental Parameters of FGK Stars.*

Advisors: Dr. Orlagh Creevey (OCA) and Dr. Caroline Soubiran (Bordeaux).

Università degli Studi di Roma "Tor Vergata"

Rome, Italy

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME) — FIRST SEMESTER

Sept. 2024 – Mar. 2025

Consortium of four universities: Università Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.

University College London

London, United Kingdom

MSCI IN PHYSICS (FIRST CLASS HONOURS)

Sept. 2018 – Jun. 2022

Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).

Thesis: *Hunting for Intermediate-Mass Black Hole Binaries.*

Research Experience

Observatoire de la Côte d'Azur

Nice, France

MASTER'S THESIS: CHARACTERISING M-DWARF STARS AS EXOPLANET HOSTS

Jan. 2026 – Sept. 2026

- Supervised by Dr. Orlagh Creevey (OCA), Dr. Ulrike Heiter (Uppsala University) and Dr. Valentina D'Orazi (Tor Vergata), within the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Deriving luminosities and radii of benchmark M-dwarf stars via spectral energy distribution (SED) fitting and bolometric flux calculations, using *Gaia* XP spectra and multi-band photometry.
- Building and testing Python pipelines for photometric and spectroscopic data comparison, integrating literature stellar parameters for cross-validation.
- Contributing to *PLATO* host-star modelling by refining fundamental parameters used in the mission's stellar characterisation framework.

Astrophysics Research Institute

Liverpool, United Kingdom

SUMMER RESEARCH PROJECT: WHITE DWARF COOLING SEQUENCES AND INFRARED COLOUR EXCESS

Jul. 2025 – Sept. 2025

- Supervised by Prof. Maurizio Salaris. Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 reflects an intrinsic IR-excess population or statistical scatter.
- Analysed JWST and HST photometric data; constructed synthetic white dwarf populations from theoretical isochrones and a Salpeter IMF, with Monte Carlo uncertainty propagation.
- Developed Python routines to compare observed and simulated colour-magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis.

Scholarships & Honours

2024–2026	Erasmus Mundus (EMJM) Scholarship , European Education and Culture Executive Agency	€33,600
2022	Summer Research Internship Scholarship , Dept. of Physics & Astronomy, UCL	£2,000
2022	"Teach Physics" Internship Award , Ogden Trust — Eastbury Community School, London	£1,400

Teaching

Eastbury Community School

Barking, United Kingdom

SUMMER PHYSICS TEACHING INTERN — OGDEN TRUST "TEACH PHYSICS" PROGRAMME

Jun. – Jul. 2022

Taught A-level and GCSE physics through interactive sessions and lab exercises; created classroom materials aligned with UK physics curricula; supported Science Club activities.

Outreach

UCL Observatory

Mill Hill, United Kingdom

DEMONSTRATOR

Oct. 2019 – Sept. 2022

Trained to operate robotic and manual telescopes (14-inch Schmidt-Cassegrain; twin 24/18-inch refractors). Led public observing sessions for school groups and members of the public; developed science communication skills across diverse audiences.

Organised outreach activities in local schools and community groups; planned and delivered science communication events including UCL's public festival *Your Universe*; produced materials translating astrophysical research for non-specialist audiences.

Service & Committees

2024–2025 **Student Representative**, Quality Evaluation Committee, Erasmus Mundus MASS Programme

Rome/Nice

Skills & Professional Memberships

Programming	Python, MATLAB, IDL
Astronomical Software	TOPCAT, SAO DS9, AstrolImageJ, VOSA, iSpec
Tools & Environments	Git, $\LaTeX$, UNIX/Linux
Professional Memberships	Institute of Physics (IOP) ; Mensa Romania ; England Handball Referee Association
Languages	English: Fluent Romanian: Native French: Intermediate